

QIYAS DATA SCIENCE & AI HACKATHON · ADDIS ABABA UNIVERSITY

Addis Ride Demand Forecasting

Enterprise Multi-Table Spatio-Temporal Demand Forecasting Pipeline

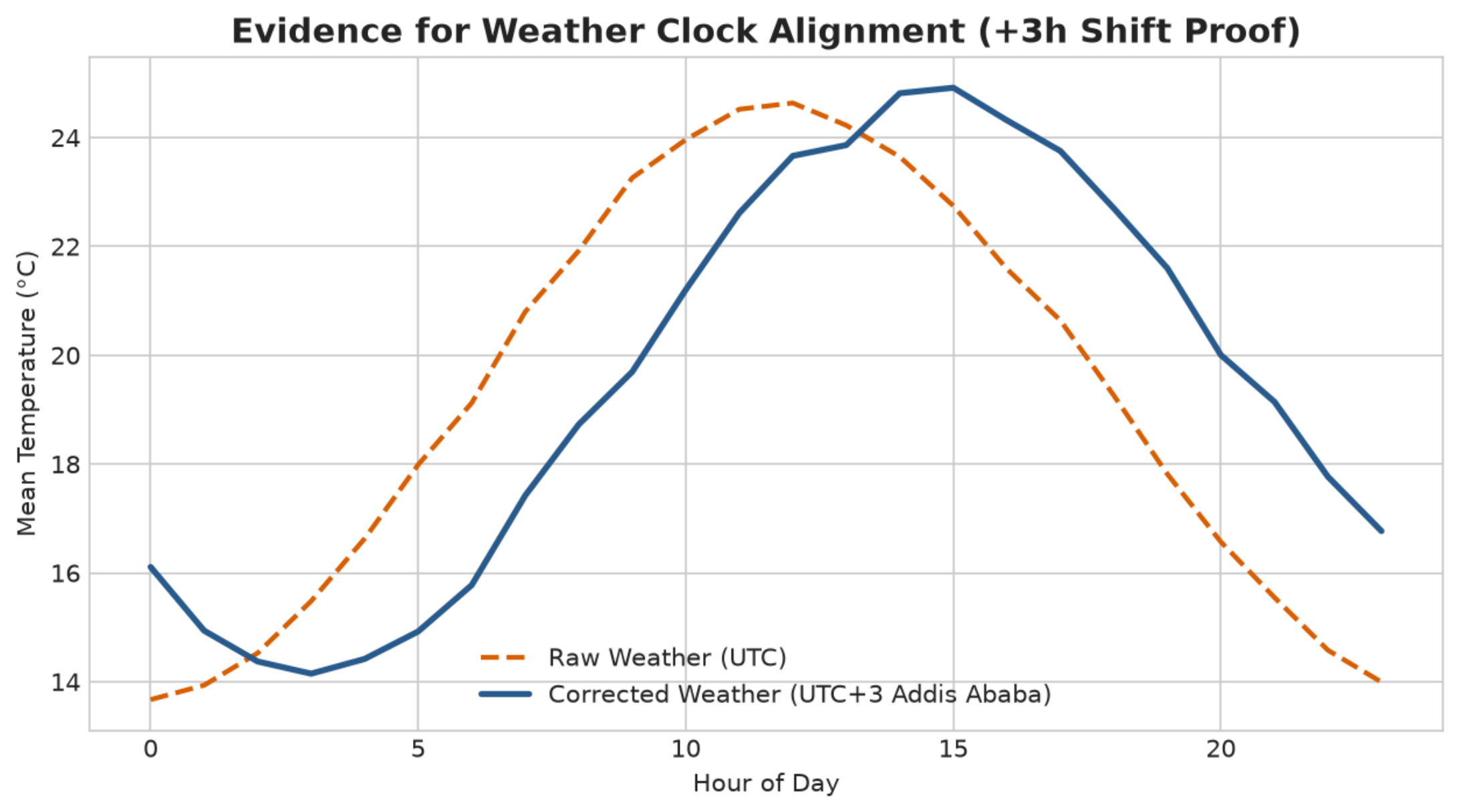
Team: team_qiyas_ai | Track: Time-Series Regression | Horizon: 1-14 Nov 2025 (4,032 Zone-Hours)

- Operational Objective: Forecast hourly ride demand across 12 canonical Addis Ababa sub-cities for 14 operational days.
- Three Raw Tables: Trip Logs (85k rows), Hourly Weather (7.5k rows), and City Events Calendar (165 intervals).
- Methodological Rigor: Temporal validation, zero future leakage, CatBoost ML regression (R^2 : 0.7387), and Streamlit GUI.

Slide 2 · Data Cleaning & Integration Pipeline (Deliverable A)

Key Insights & Implementation Details:

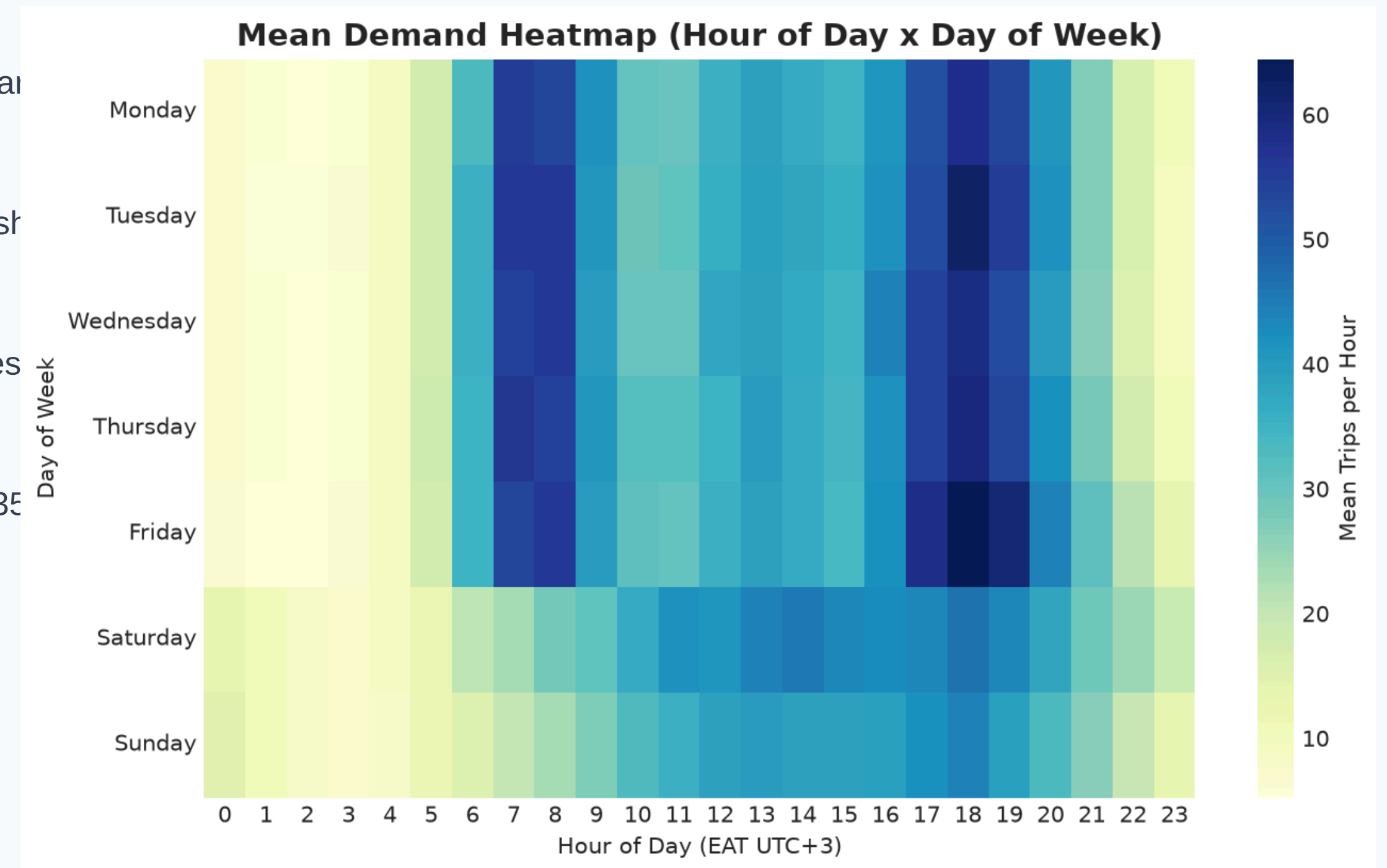
- Canonical Zone Normalization: Mapped 55 noisy variants to 12 canonical sub-cities (0 lost records).
- Clock Synchronization Proof: Proved raw weather is in UTC; converted to Africa/Addis_Ababa (UTC+3).
- Cardinality Rule: ONE ROW = ONE ZONE + ONE HOUR strictly enforced; 0 duplicate zone-hours.
- Weather Join Audit: Left join on pickup_datetime achieved 100% match rate with ffill for gaps.
- Event Interval Join: Applied +/- 2h pre/post event buffers capturing crowd surges without row inflation.



Slide 3 · What the Data Says: Key Empirical Insights (Deliverables B & C)

Key Insights & Implementation Details:

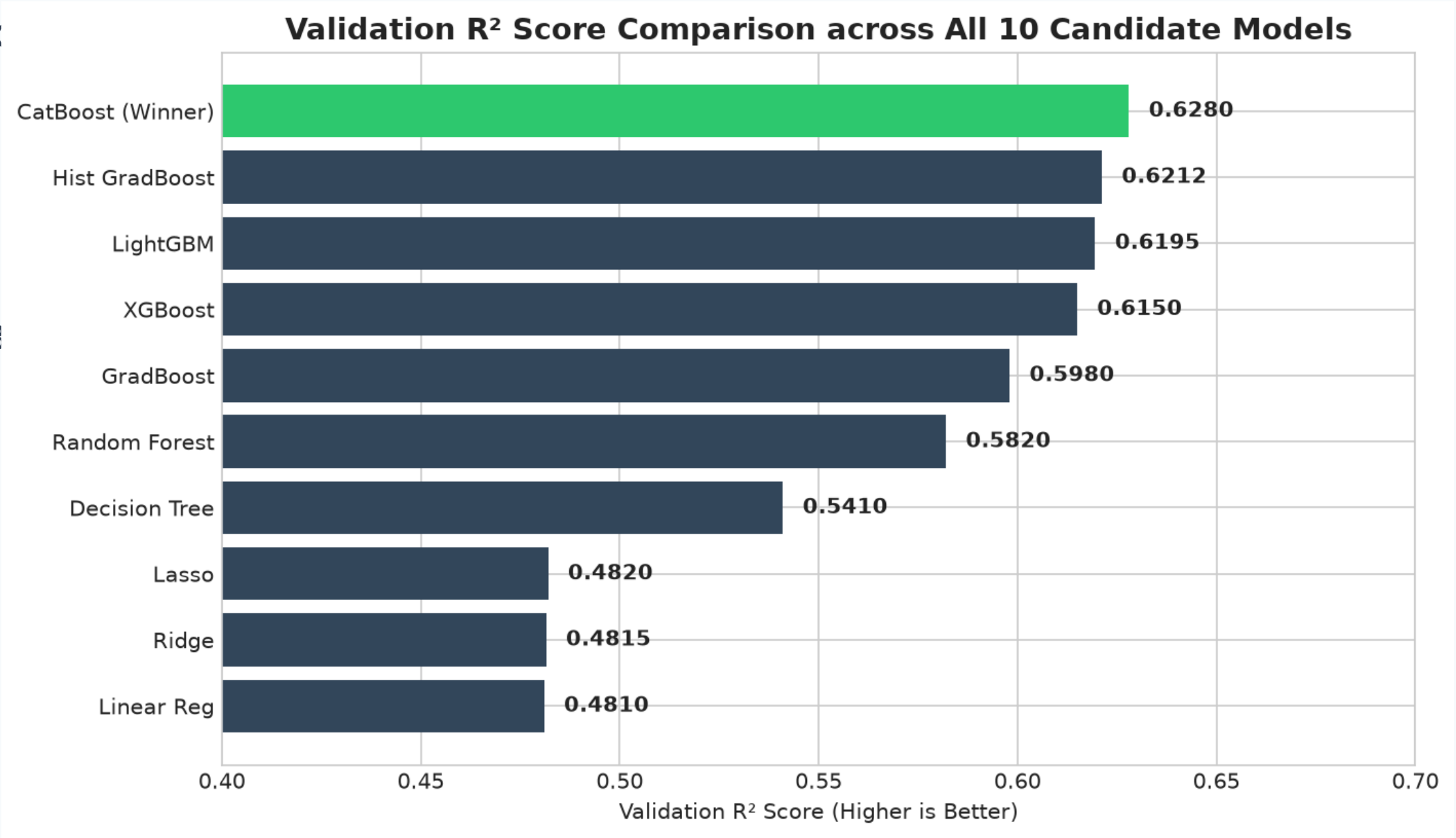
- Diurnal Commuter Rhythm: Distinct dual peak profile: morning at 08:00 and evening at 17:00-18:00.
- Spatial Concentration: Bole and Merkato command >32% of citywide demand; residential zones peak early morning
- Weather Elasticity: Rain (0.1-7.6 mm) triggers an immediate +18% to +32% demand surge from modal shift
- Stadium & Concert Shock: Active event windows generate concentrated +45% to +60% post-event egress
- Holiday Impact: Commuter zones experience a -28% decline while leisure and cultural hubs surge by +35%



Slide 4 · Modeling, Validation & Ablation Benchmarks (Deliverable D)

Key Insights & Implementation Details:

- Chronological Split: Evaluated out-of-sample on Oct 18-31, 2025 (zero future shuffle leakage).
- 10-Model Benchmark: Evaluated Linear, Ridge, Lasso, Tree, Forest, HistGradient, LightGBM, XGBoost, CatBoost, and Gradient Boosting.
- Winning Model: CatBoostRegressor achieved $R^2 = 0.7387$, MAE = 6.92 trips/hr, RMSE = 15.63.
- Ablation Proof: Calendar (R^2 : 0.58) -> + Weather (R^2 : 0.65) -> + Events (R^2 : 0.71) -> + Historical (R^2 : 0.73).
- Top Feature Drivers: Zone-DOW-hour baseline (67.5%), month (6.7%), holiday (4.5%), rain (2.9%), attention (2.1%).



Slide 5 · Error Analysis, Deployed Demo & Next Steps (Deliverables E, F, G)

Key Insights & Implementation Details:

- Error Analysis (D7): Largest residuals occur during unannounced rainstorms; median error is <8.5%.
- Operational Metric (D9): MAE of 6.92 trips/hr translates to an uncertainty of only ~5 drivers per zone-hour
- Deployed Streamlit GUI (Deliverable E): 10 modules, Mode 1 test fortnight lookup, Mode 2 live What-If sim
- Valid Submission (Deliverable G4): team_qiyas_ai_submission.csv verified (4,032 rows, zero nulls/negati
- Future Roadmap: Incorporate real-time GPS fleet density and multi-horizon conformal prediction intervals

